

AI Usage Note — AlgoChowk Quant Engineer Assignment

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Tools used

- ChatGPT for research-design review, statistical-method discussion, code review, edge-case identification, documentation and presentation drafting.

How AI was used

- Converted the assignment's broad hypothesis into a reproducible event-study specification.
- Reviewed potential look-ahead bias and entry-timing choices.
- Suggested baseline comparisons, bootstrap confidence intervals, overlap handling, robustness checks and chronological OOS validation.
- Helped structure the Python research engine and submission documents.

My own decisions

- Primary event threshold: -2%.
- Primary holding period: 5 trading days.
- Entry: next trading-day open.
- Closely occurring events: suppress new signals until the 5-day holding window ends.
- Development/OOS split: 2015–2020 / 2021–2025.
- Friction proxy: 20 bps round trip.
- The primary specification was frozen before interpreting the final results.

Suggestions changed or challenged

- I did not optimize the threshold to maximize historical returns.
- I did not select the strongest robustness cell as the final strategy.
- I treated the backtest as an extension of the research rather than the objective.
- I retained a negative conclusion where the evidence did not support the original hypothesis.

Incorrect / risky suggestions identified

- A backtest that enters at the event-day close would introduce look-ahead because the event is defined using that day's close.
- Treating every closely spaced event as independent would understate dependence.
- Choosing the best threshold after inspecting all historical results would create post-hoc selection/data-snooping risk.

What I learned

- Event studies need a precise signal timestamp, not just a price condition.
- A positive average return is not evidence of an edge unless it is compared with an appropriate baseline.
- Robustness and OOS validation matter more than a high in-sample return.
- Statistical significance and economic significance are different questions.